Connector Economies
变量构造目录与使用指南

内部严格审查材料

variable_catalog_v0.2.0

由 F:/working_paper_wenxiao/trade_welfare/docs/variable_construction_catalog.md
可重复生成

用途：合作者复核数据来源、处理规则、变量定义、验证结果和解释边界。

目录

Material Passport 4
0. 结论和使用方式 4
1. 状态、粒度和可加总性代码 4
1.1 状态 . . . 4
1.2 粒度代码 . . . 5
1.3 可加总性代码 . . . 5
2. 通用键、标签和 provenance 变量 5
3. M2 已验证的政策变量 6
4. M2 已验证的贸易金额和 Census 变量 7
4.1 Connector trade . . . 7
4.2 Census . . . 7
5. M3.1 connector-sector-year 基础聚合变量 7
5.1 金额和覆盖 . . . 8
5.2 sector-level treatment . . . 9
5.3 标签、版本和不可加总 flags . . . 10
5.4 M3.1 验证结论和解释边界 . . . 10
6. M3.2 时间窗口变量 10
6.1 zero 和 small-base flags . . . 11
7. M3.3-M3.6 贸易重组和排名输入变量 11
7.1 Import replacement . . . 11
7.2 Source-to-connector input growth . . . 12
7.3 Product-level co-movement . . . 12
7.4 Exposure overlap . . . 12
7.5 稳定性和规模 . . . 12
8. M4 production-side 标准化变量 12
8.1 通用 long schema . . . 13
8.2 ICIO metrics . . . 13
8.3 AMNE metrics . . . 13
8.4 BEA MNE metrics . . . 14
8.5 BEA FDI metrics . . . 14
8.6 Mexico FDI metrics . . . 14
9. M4-M5 production changes 和覆盖变量 14
10. M6 connector ranking 15
10.1 组件 . . . 15
10.2 至少两种排名方案 . . . 15
10.3 Ranking outputs . . . 16

11. M6 rerouting facts 和 production validation 16
11.1 Rerouting classification . . . 16
11.2 Production validation . . . 16
12. Structural moments 和 Scenario A/B/C 17
12.1 Trade-cost candidates . . . 17
12.2 MP/production targets . . . 17
12.3 Structural moment table . . . 17
13. Controls、fixed effects 和 robustness 变量 18
14. 不允许的变量和解释 18
15. Stata 文件、主键和验证合同 19
15.1 计划交付 . . . 19
15.2 Storage types . . . 19
15.3 必须运行的 Stata 检查 . . . 19
16. 每个变量完成后的登记模板 20
17. 当前施工顺序 20
18. 版本记录 20

Material Passport

- Origin Skill: stata-latex-research
- Origin Date: 2026-07-27
- Verification Status: VERIFIED_THROUGH_M3.1
- Version Label: variable_catalog_v0.2.0
- Current Data Stage: M0-M3.1 completed
- Next Construction Stage: M3.2 not started
- Canonical Format: Parquet
- Collaborator Format: Stata .dta

0. 结论和使用方式

从 M3.1 开始，变量构造应以本目录为变量合同。它回答六个问题：

1. 变量在哪一层数据上构造；
2. 变量的公式、分母、时间窗口和单位是什么；
3. 变量用于 connector ranking、rerouting、production validation 还是 structural moments；
4. 零值、缺失、负值、small base、suppression 和 confidentiality 如何处理；
5. 变量能否求和、能否跨 connector 比较、能否进入回归或模型；
6. 导出 Stata 时使用什么名称、storage type、主键和验证命令。

本目录同时包含三类变量：

- EXISTING_VALIDATED：M2 或 M3.1 已存在且 validation PASS；
- PLANNED_M3 至 PLANNED_M6：已经定义但尚未构造，必须在对应阶段完成后补充实际 output hash 和 validation；
- STRUCTURAL_TARGET_ONLY：只能作为模型校准目标或输入候选，不能直接解释为已经识别的结构参数。

凡是本目录未登记的正式变量，不应直接写入最终 processed data。临时探索变量可以存在于 tmp/ 或交互式 Stata session，但进入正式输出前必须补齐本目录。

本目录与 docs/data_processing_audit_trail.md 构成同一套审查材料。本目录不重复数据来源和完整处理历史，而是引用审计档案中的输入、输出和 validation 证据；审计档案也不能替代这里的变量级公式、用途和解释边界。M3-M7 任一阶段完成时，两份 Markdown、两份 PDF 和对应 build summary 必须同步更新。

1. 状态、粒度和可加总性代码

1.1 状态

状态	含义
EXISTING_VALIDATED	已在 processed Parquet 中存在，主键和 validation 已通过
PLANNED_M3	M3.2-M3.6 的窗口、增长、replacement 和 ranking inputs 尚未构造
PLANNED_M4	production-side standardized long table 时构造
PLANNED_M5	trade 和 production evidence 合并时构造
PLANNED_M6	ranking、rerouting、validation 和 structural moments 时构造
STRUCTURAL_TARGET_ONLY	观测 moment 或 candidate shock，不等于已识别结构参数
PROHIBITED	明确禁止构造或禁止解释的变量

1.2 粒度代码

Grain ID	唯一键	主要文件
G_PRODUCT_YEAR	analysis_hs6, year	product treatment
G_CONN_PRODUCT_YEAR	connector_iso3, analysis_hs6, year	connector trade
G_ORIGIN_PRODUCT_MONTH	origin_iso3, analysis_hs6, year, month	Census monthly
G_CONN_SECTOR_YEAR	connector_iso3, analysis_sector, year	M3 trade-sector
G_CONN_SECTOR_WINDOW	connector_iso3, analysis_sector, window_id	M3 window summaries
G_PROD_LONG	country_iso3, analysis_sector, year, source_name, metric_name，必要时增加 direction/parent	M4 production long
G_EVIDENCE	connector_iso3, analysis_sector, year	M5 integrated evidence
G_RANK	connector_iso3, analysis_sector, ranking_spec	M6 ranking
G_MOMENT	moment_id, scenario	M6 structural moments

1.3 可加总性代码

Code	规则
ADD_WITHIN_KEYS	可以在不跨越声明维度时求和，例如产品金额聚合到 sector
NONADDITIVE_RATE	rate/share/exposure 不能直接求和，必须重算分子分母
NONADDITIVE_REPEAT	为比较而重复，绝不能跨重复维度求和
STOCK	时点/期末 stock，不能跨年求和
FLOW	期间 flow，可在不重叠期间内求和
LEVEL	level，可比较或求差，但通常不跨单位/来源相加
FLAG	逻辑状态，不作金额加总

2. 通用键、标签和 provenance 变量

这些字段必须尽可能贯穿 M3-M7。它们不是辅助杂项，而是合作者复核 join 和解释边界的基础。

Canonical name	Stata name	Type	Grain	用途	状态
connector_iso3	conn_iso3	str3	connector grains	connector 国家键	existing/planned
country_iso3	country_iso3	str3	production	production host/country 键	planned M4
analysis_hs6	hs6	str6	product grains	保留 leading zeros 的产品键	existing
analysis_sector	ana_sector	str32	sector grains	第一阶段共同经验 sector	existing
year	year	int	annual grains	calendar year	existing
month	month	byte	monthly	1-12	existing
window_id	window_id	str8	window	pre/shock/post	planned M3
source_name	source	str20	production long	数据来源区分	planned M4
metric_name	metric	str32	production long	指标区分	planned M4
unit	unit	str32	all long outputs	单位，禁止跨单位相加	planned M4
mapping_version	map_ver	str32	all	concordance 版本	planned
build_version	build_ver	str32	all	builder/output schema 版本	existing/planned
source_row_count	src_n	long	all	聚合 cell 的原始行数	existing/planned
source_file_count	srcfile_n	int	all	provenance 完整性	existing/planned
mapping_status	map_status	str32	all	mapped/aggregate/residual/ambiguous 等	existing/planned
is_analysis_eligible	ana_ok	byte	all	是否进入第一阶段主分析	existing/planned
is_proxy	is_proxy	byte	all	TWN S19 或其他 proxy	existing/planned
is_aggregate	is_agg	byte	production/country	aggregate geography/sector	existing/planned
is_residual	is_resid	byte	country	Mexico residual 等	existing/planned

Canonical name	Stata name	Type	Grain	用途	状态
suppressed_flag	suppressed	byte	production	BEA suppressed	planned M4
confidential_flag	confidentl	byte	production	Mexico confidential	existing/planned
allocated_flag	allocated	byte	production	AMNE allocated metric	planned M4
nonadditive_flag	nonadd	byte	all	repeated/rate/stock 不可加总	planned

Stata 中所有逻辑变量使用 numeric byte 0/1，并附 value label。不能把 Python boolean 转成字符串 "True"/"False"。

3. M2 已验证的政策变量

文件：data/processed/harmonized/product_treatment_hs6_year.parquet。粒度 G_PRODUCT_YEAR。

Canonical name	Stata name	定义/单位	可加总性	用途	状态
total_us_import_value_2017	usimp17	2017 total U.S. imports，USD	level；映射后可加总	treatment 分母	existing validated
china_import_value_2017	chnimp17	2017 China imports，USD	level；映射后可加总	China share 和 weights	existing validated
china_import_share_2017	chnshr17	chnimp17/usimp17，fraction	nonadditive rate	exposure 分母结构	existing validated
tariff_rate_china_weighted	trf_chw	source HS17 rate 按 China imports 加权	nonadditive rate	sensitivity/provenance	existing validated
tariff_rate_total_import_weighted	trf_usw	source rate 按 total imports 加权	nonadditive rate	zero-China fallback	existing validated
tariff_rate	trf_rate	最终 statutory rate，fraction	nonadditive rate	policy intensity	existing validated
tariff_exposure	trf_expo	sum(China imports × rate)/sum(total imports)	nonadditive rate	主 gross exposure	existing validated
tariffed_product	tariffed	任一 mapped source code rate > 0	flag	treatment indicator	existing validated
source_hs17_code_count	hs17_n	汇入 analysis product 的 HS17 数量	count	audit many-to-one	existing validated
policy_source_observed_count	polobs_n	legal panel 中 observed source codes	count	区分 not-listed 与 missing	existing validated
source_tariffed_hs17_count	trfhs17_n	positive-rate source codes	count	treatment composition	existing validated
tariff_rate_max_source	trf_max	mapped sources 最大法定 rate	nonadditive rate	denominator=0 fallback	existing validated
policy_mapping_status	pol_map	weighted/no denominator/no HS17 weight	category	sample/filter/audit	existing validated
historical_exclusions_applied	hist_excl	当前恒为 0	flag	防止误称 realized duties	existing validated
four_year_review_2024_included	rev24_in	当前恒为 0	flag	防止误并 2024 review	existing validated

禁止变量：

Prohibited name/concept	原因
realized_tariff_paid from current panel	当前数据未重建 customs-entry realized duties
tariff_exposure_with_exclusions	historical exclusions 尚未应用
2024_review_hs6 by first-six-digit truncation	modern HTS scope 和 historical HS6 不可直接等同

4. M2 已验证的贸易金额和 Census 变量

4.1 Connector trade

文件：connector_trade_hs6_year.parquet。粒度 G_CONN_PRODUCT_YEAR。

Canonical name	Stata name	单位	可加总性	用途	状态
china_to_usa_value_usd	cn_us_val	current USD	NONADDITIVE_REPEAT across connector	共同 China-USA benchmark	existing validated
china_to_connector_value_usd	cn_conn_val	current USD	add across products, not connectors unless desired	China input leg	existing validated
east_asia_to_connector_value_usd	ea_conn_val	current USD	add across products	East Asia input leg	existing validated
connector_to_usa_value_usd	conn_us_val	current USD	add across products/connectors with clear scope	connector export leg	existing validated
*_row_count	shortened *_n	count	additive count	区分 true zero 与 source rows	existing validated
east_asia_self_trade_excluded	ea_selfx	flag	flag	KOR/TWN self-trade rule	existing validated
baci_absence_treated_as_zero	baci_zrule	flag	flag	complete-grid zero rule	existing validated
china_to_usa_common_benchmark	cnus_common	flag	flag	repeated warning	existing validated
treatment_repeated_across_connectors	trf_repeat	flag	flag	non-additive warning	existing validated

4.2 Census

月度 grain G_ORIGIN_PRODUCT_MONTH；年度 grain origin_iso3, analysis_hs6, year。

Canonical name	Stata name	单位/公式	用途	状态
import_value_usd	imp_val	current USD	主进口金额	existing validated
customs_value	cust_val	current USD	customs valuation check	existing validated
duties	duties	current USD	collected duties 描述，不等于 Section 301-only duties	existing validated
dutiable_value_usd	dutiable	current USD	dutiable base	existing validated
observed_month_count	month_n	0-12	年度 cell 的月度覆盖	existing validated
calendar_source_months_complete	cal12_ok	flag	允许 sparse absence -> zero 的前提	existing validated
quantity_usable	qty_ok	当前恒为 0	禁止 quantity analysis	existing validated
is_all_origin_denominator	allorig	当前恒为 0	防止误作全来源分母	existing validated

duties/import_value 可以作为描述性 average duty ratio，但不能命名为 Section 301 realized rate，除非另行识别 tariff program 和 entry-level scope。

5. M3.1 connector-sector-year 基础聚合变量

状态：EXISTING_VALIDATED。

正式文件：data/processed/harmonized/connector_trade_sector_year.parquet。粒度 G_CONN_SECTOR_YEAR，唯一键为 connector_iso3, analysis_sector, year。

构建证据：

- builder：code/data_processing/build_connector_trade_sector_panel.py；
- 冻结配置：configs/connector_trade_sector_rules.yaml；
- 输入：743,886 行 connector_trade_hs6_year.parquet 与 5,202 行 product_sector_crosswalk.parquet；

- 输出：1,001 行、70 列，13 connectors × 7 sectors × 2014-2024；
- 输出 SHA256：95e547267892c4bc883788542f0e59ba284441b42f4ce1d379aea71a2f978c8c；
- validation：data/validation/connector_trade_sector_panel_validation.json 与 .md，状态 PASS；
- 配置 SHA256：6a313c01d9bfad6e6e6a2cb508bd787f35bdb0eddeb6e4f491a86d61bdd9ee9d；
- 确定性检查：项目 F 盘临时目录第二次构建的 Parquet SHA256 与正式输出完全一致，临时目录已清除；
- Stata 接口：70 个 alias 全部登记、互不重复、最长 14 个字符，均不超过 Stata 的 32 字符限制。本阶段不提前生成 .dta。

5.1 金额和覆盖

先用 analysis_hs6 把 M2 panel many-to-one 连接到 canonical product-sector crosswalk。只有 canonical is_analysis_sector_eligible=true 且 sector_mapping_weight=1 的产品，才在 connector_iso3, analysis_sector, year 内求和。M2 内嵌 sector 字段只用于一致性检查，不能取代 canonical crosswalk。

Canonical name	Stata name	公式	可加总性	用途	状态
china_to_usa_value_usd	cn_us_val	sum_h eligible china_to_usa_value_usd	NONADDITIVE_REPEAT across connector	China decline benchmark	EXISTING_VALIDATED
china_to_connector_value_usd	cn_conn_val	sum_h eligible china_to_connector_value_usd	FLOW	China input leg	EXISTING_VALIDATED
east_asia_to_connector_value_usd	ea_conn_val	sum_h eligible east_asia_to_connector_value_usd	FLOW	broader source input leg	EXISTING_VALIDATED
connector_to_usa_value_usd	conn_us_val	sum_h eligible connector_to_usa_value_usd	FLOW	connector export leg	EXISTING_VALIDATED
sector_eligible_product_count	sec_elig_n	sector 内 distinct eligible HS6	count	sector composition 与 denominator	EXISTING_VALIDATED

四条流各有独立 diagnostics；<flow> 分别是 china_to_connector、east_asia_to_connector、connector_to_usa 和 china_to_usa：

Canonical pattern	Stata aliases	公式和含义	状态
<flow>_value_usd_row_count	cn_conn_src_n, ea_conn_src_n, conn_us_src_n, cn_us_src_n	eligible products 的 M2 source-row count 之和	EXISTING_VALIDATED
<flow>_positive_product_count	cn_conn_pos_n, ea_conn_pos_n, conn_us_pos_n, cn_us_pos_n	sector 内 <flow>_value_usd > 0 的 eligible HS6 数	EXISTING_VALIDATED
<flow>_zero_source_product_count	cn_conn_zsrc_n, ea_conn_zsrc_n, conn_us_zsrc_n, cn_us_zsrc_n	sector 内 M2 <flow>_row_count = 0 的 eligible HS6 数；保留 source absence，不等同 observed zero	EXISTING_VALIDATED

excluded、outside-manufacturing 和 residual products 没有可用于 sector sums 的有效 sector weight，因此 coverage 在 connector_iso3, year 粒度计算后跨 7 个 sector 重复。所有 coverage 字段都带 coverage_repeated_across_sectors=true，不能跨 sector 求和。

Canonical name	Stata name	公式	可加总性	状态
excluded_product_count	excl_prod_n	connector-year 内 965 个非 eligible 产品的 distinct count	NONADDITIVE_REPEAT across sector	EXISTING_VALIDATED
china_to_connector_excluded_value_usd	cn_conn_excl	非 eligible 产品的 China→connector value sum	NONADDITIVE_REPEAT across sector	EXISTING_VALIDATED
china_to_connector_included_value_share	cn_conn_incl_sh	eligible / full-universe China→connector value	NONADDITIVE_RATE	EXISTING_VALIDATED
east_asia_to_connector_excluded_value_usd	ea_conn_excl	非 eligible 产品的 East Asia→connector value sum	NONADDITIVE_REPEAT across sector	EXISTING_VALIDATED
east_asia_to_connector_included_value_share	ea_conn_incl_sh	eligible / full-universe East Asia→connector value	NONADDITIVE_RATE	EXISTING_VALIDATED
connector_to_usa_excluded_value_usd	conn_us_excl	非 eligible 产品的 connector→USA value sum	NONADDITIVE_REPEAT across sector	EXISTING_VALIDATED
connector_to_usa_included_value_share	conn_us_incl_sh	eligible / full-universe connector→USA value	NONADDITIVE_RATE	EXISTING_VALIDATED

Canonical name	Stata name	公式	可加总性	状态
china_to_usa_excluded_value_usd	cn_us_excl	非 eligible 产品的 China→USA value sum；同时跨 connector 重复	NONADDITIVE_REPEAT	EXISTING_VALIDATED
china_to_usa_included_value_share	cn_us_inclsh	eligible / full-universe China→USA value；同时跨 connector 重复	NONADDITIVE_RATE	EXISTING_VALIDATED

5.2 sector-level treatment

先对 analysis_hs6, year 去除 connector 复制，再在 sector-year 重算分子和分母。Policy rate/exposure 不能按产品 rate 直接求和。

Canonical name	Stata name	公式	用途	状态
sector_total_us_import_value_2017	sec_usimp17	sum_h usimp17_h	transparent denominator	EXISTING_VALIDATED
sector_china_import_value_2017	sec_cnimp17	sum_h chnimp17_h	China weight denominator	EXISTING_VALIDATED
sector_tariff_exposure_numerator_usd	sec_trf_num	sum_h(chnimp17_h × rate_h,t)	common transparent numerator	EXISTING_VALIDATED
sector_tariff_rate_china_weighted	sec_trf_cw	sec_trf_num / sec_cnimp17	法定 rate intensity	EXISTING_VALIDATED
sector_tariff_exposure	sec_expo	sec_trf_num / sec_usimp17	主 gross policy exposure	EXISTING_VALIDATED
sector_china_import_share_2017	sec_cnshr17	sec_cnimp17 / sec_usimp17	baseline China dependence	EXISTING_VALIDATED
sector_treatment_product_count	sec_trt_n	distinct eligible HS6	policy count denominator	EXISTING_VALIDATED
policy_denominator_observed_product_count	pol_denobs_n	usimp17_h > 0 且非缺失的 eligible HS6 数	denominator audit	EXISTING_VALIDATED
tariffed_product_count	trf_prod_n	tariffed_product=true 的 eligible HS6 数	breadth numerator	EXISTING_VALIDATED
tariffed_china_import_value_2017	trf_cnimp17	tariffed eligible HS6 的 China imports sum	China treatment mass	EXISTING_VALIDATED
tariffed_total_us_import_value_2017	trf_usimp17	tariffed eligible HS6 的 total U.S. imports sum	total-import treatment mass	EXISTING_VALIDATED
tariffed_product_share_count	trf_psh_n	trf_prod_n / sec_trt_n	breadth	EXISTING_VALIDATED
tariffed_product_share_china_value	trf_psh_c	trf_cnimp17 / sec_cnimp17	China treatment mass share	EXISTING_VALIDATED
tariffed_product_share_us_value	trf_psh_u	trf_usimp17 / sec_usimp17	total-import treatment mass share	EXISTING_VALIDATED

缺少 denominator 或 HS2017 mapping 的产品不静默转成普通零 denominator：

Canonical name	Stata name	公式/含义	状态
treatment_missing_denominator_product_count	trt_missden_n	usimp17_h 缺失或不大于 0 的 eligible HS6 数	EXISTING_VALIDATED
treatment_zero_denominator_product_count	trt_zeroden_n	policy_mapping_status=mapped_hs17_no_2017_import_denominator 的 HS6 数；全样本 17 个 distinct products	EXISTING_VALIDATED
treatment_no_hs17_mapping_product_count	trt_nohs17_n	policy_mapping_status=no_hs17_baci_weight_mapping 的 HS6 数；全样本 11 个 distinct products	EXISTING_VALIDATED
treatment_missing_denominator_product_share	trt_miss_sh	missing denominator count / sec_trt_n	EXISTING_VALIDATED
treatment_zero_denominator_product_share	trt_zero_sh	zero denominator count / sec_trt_n	EXISTING_VALIDATED
treatment_no_hs17_mapping_product_share	trt_nohs17_sh	no-HS2017-mapping count / sec_trt_n	EXISTING_VALIDATED

主回归/排名应优先使用 sector_tariff_exposure，同时把 weighted rate 和 tariffed shares 作为 robustness，不把它们互相替代。

5.3 标签、版本和不可加总 flags

Canonical name	Stata name	内容/规则	状态
connector_group	conn_group	core/comparator 分组	EXISTING_VALIDATED
connector_baci_code	conn_baci	BACI reporter/partner code	EXISTING_VALIDATED
connector_is_proxy	is_proxy	TWN 使用 BACI S19 proxy；TWN 的 77 行全部为 1	EXISTING_VALIDATED
analysis_sector_status	sec_status	canonical sector status	EXISTING_VALIDATED
trade_value_unit	val_unit	current USD	EXISTING_VALIDATED
treatment_concept	trt_concept	gross statutory initial actions	EXISTING_VALIDATED
historical_exclusions_applied	hist_excl	恒为 0	EXISTING_VALIDATED
four_year_review_2024_included	rev24_in	恒为 0	EXISTING_VALIDATED
review_2024_treatment	rev24_trt	separate exact-HTS extension	EXISTING_VALIDATED
treatment_version	trt_ver	M2 treatment version	EXISTING_VALIDATED
is_analysis_sector_eligible	ana_ok	本输出恒为 1	EXISTING_VALIDATED
sector_mapping_weight	sec_wt	本输出恒为 1	EXISTING_VALIDATED
baci_absence_treated_as_zero	baci_zrule	complete-grid BACI absence rule	EXISTING_VALIDATED
quantity_included	qty_in	恒为 0；本输出不含 quantity	EXISTING_VALIDATED
china_to_usa_common_benchmark	cnus_common	China→USA benchmark 跨 connector 重复	EXISTING_VALIDATED
treatment_repeated_across_connectors	trt_repeat	sector treatment 跨 connector 重复	EXISTING_VALIDATED
coverage_repeated_across_sectors	cov_repeat	connector-year coverage 跨 sector 重复	EXISTING_VALIDATED
common_benchmarks_not_additive_across_connectors	bm_nonadd	禁止跨 connector 求和共同 benchmark	EXISTING_VALIDATED
product_mapping_version	prod_mapver	canonical product-sector mapping version	EXISTING_VALIDATED
m3_sector_aggregation_version	m3_aggver	connector_trade_sector_rules_v1	EXISTING_VALIDATED
schema_version	schema_ver	connector_trade_sector_year_v1	EXISTING_VALIDATED

5.4 M3.1 验证结论和解释边界

- 两个输入 SHA256 与冻结配置一致；输入、crosswalk 和输出主键重复均为 0；全部 HS6 成功匹配，M2 内嵌 sector 字段与 canonical crosswalk 不一致数为 0。
- eligible HS6 数固定为 autos 64、electrical equipment 177、electronics 252、machinery 467、medical devices 32、other manufacturing 2,422、textiles/apparel 823。
- 四条 flow 的 product-level eligible totals 与 sector totals 差额均为 USD 0；eligible + excluded = full universe 的 connector-year 最大差额均为 USD 0。
- 九个 policy rate/share 字段独立公式复算最大绝对误差均为 0，小于 1e-12。
- China→USA benchmark、policy 和 coverage 的重复一致性不一致数均为 0。
- 当前输出只含年度 level、coverage 和 sector treatment。它不含 pre/shock/post window、growth、small-base、replacement ratio、ranking、production-side 或 structural moments。

6. M3.2 时间窗口变量

不同窗口长度不相等：pre 4 年、shock 2 年、post 5 年。因此 level comparison 使用年度平均值，不使用窗口总和。

对任意非负年度 flow x_t:

```
x_pre  = mean(x_t, 2014-2017)
x_shk  = mean(x_t, 2018-2019)
x_post = mean(x_t, 2020-2024)

d_x_post_pre = x_post - x_pre
g_dh_x_post_pre = 2 * (x_post - x_pre) / (x_post + x_pre)
g_log_x_post_pre = ln(x_post) - ln(x_pre), only if both > 0
```

g_dh 是推荐 primary growth：无量纲、允许一端为 0、范围 [-2,2]。g_log 只用于 positive-base robustness。禁止把 undefined log growth 自动设为 0。

每条贸易腿构造：

Family	Pre level	Shock level	Post level	Level change	DH growth	Positive-base log growth
China -> USA	cnus_pre	cnus_shk	cnus_post	d_cnus	gdh_cnus	gln_cnus
China -> connector	cncn_pre	cncn_shk	cncn_post	d_cncn	gdh_cncn	gln_cncn
East Asia -> connector	eacn_pre	eacn_shk	eacn_post	d_eacn	gdh_eacn	gln_eacn
connector -> USA	cnusx_pre	cnusx_shk	cnusx_post	d_cnusx	gdh_cnusx	gln_cnusx

Stata 名称必须在生成前再次检查不超过 32 characters，并在 variable labels 写出完整方向。

6.1 zero 和 small-base flags

Canonical name	Stata name	定义	用途
pre_zero_flag	pre_zero	pre annual mean = 0	区分新贸易与连续增长
post_zero_flag	post_zero	post annual mean = 0	区分退出
small_base_flag	smallbase	pre level 低于 M3 config 明确阈值	防止 ratio/rank 爆炸
log_growth_defined	logg_ok	pre>0 and post>0	log sample
years_observed_pre	pre_yn	pre 窗口实际 years	coverage
years_observed_post	post_yn	post 窗口实际 years	coverage

small-base threshold 必须写入 M3 config，至少报告：

- absolute USD threshold；
- sector-relative percentile threshold；
- 不使用 threshold、只用 DH growth 的结果。

推荐主排名使用 level change + DH growth，不依赖单一百分比增长。

7. M3.3-M3.6 贸易重组和排名输入变量

7.1 Import replacement

```
china_us_decline = max(cnus_pre - cnus_post, 0)
connector_us_gain = cnusx_post - cnusx_pre
replacement_ratio = connector_us_gain / china_us_decline
```

replacement_ratio 只有在 china_us_decline 大于明确的 denominator threshold 时定义；否则 missing。主结果同时保留 level gain 和 decline，不能只报告 ratio。

Canonical name	Stata name	用途	状态
china_us_decline_usd	cnus_drop	China displacement level	planned M3
connector_us_gain_usd	conn_gain	connector gain level	planned M3
replacement_ratio	repl_ratio	connector gain 相对 China decline	planned M3
replacement_ratio_defined	repl_ok	denominator 是否可用	planned M3
replacement_capped_for_rank	repl_cap	仅 ranking 的 winsorized/capped version	planned M3

7.2 Source-to-connector input growth

Canonical name	Stata name	定义
china_input_gain_usd	cnin_gain	China -> connector post-pre level change
east_asia_input_gain_usd	eain_gain	East Asia -> connector post-pre level change
china_input_dh_growth	gdh_cnin	China input DH growth
east_asia_input_dh_growth	gdh_eain	East Asia input DH growth
source_export_ratio_pre	srcx_pre	source-to-connector / connector-to-USA，pre
source_export_ratio_post	srcx_post	同一 ratio，post
source_export_ratio_change	d_srcx	post-pre

ratio denominator 为 0 时保持 missing；不能用 arbitrary epsilon 生成正式值。若 ranking 需要 finite score，先用 percentile of level/growth components，而不是偷偷给 ratio 加常数。

7.3 Product-level co-movement

在 connector-sector 内先对每个 HS6 计算 pre/post changes，再聚合：

Canonical name	Stata name	公式/用途
both_legs_positive_product_share	both_psh	China/East Asia input gain >0 且 connector-US gain >0 的 HS6 share
both_legs_positive_value_share	both_vsh	上述产品按 pre/post trade value 加权 share
china_decline_connector_gain_share	rerout_psh	China-US decline 且 connector-US gain 的 product share
triple_condition_product_share	triple_psh	China-US decline、source input gain、connector-US gain 同时成立
product_change_correlation	prod_corr	source-input change 与 connector-export change 的 HS6 correlation
weighted_product_change_correlation	wprod_cor	以 pre China-US 或 connector trade weights 加权

相关系数至少要求有效 products 达到 config 中的 minimum count；否则 missing + corr_low_n_flag=1。

7.4 Exposure overlap

Canonical name	Stata name	公式
connector_export_tariff_overlap_pre	ovlp_pre	pre connector-US exports 中 tariffed/exposed product 的 value share
connector_export_tariff_overlap_post	ovlp_post	post share
connector_export_exposure_weighted	ovlp_expo	connector export weights 下的 product tariff exposure
gain_in_high_exposure_products_share	gain_hiexp	export gain 来自 high-exposure products 的份额

“high exposure” threshold 必须在 config 中固定，例如 top quartile 或 positive exposure；两种定义均应报告。

7.5 稳定性和规模

Canonical name	Stata name	用途
pre_trade_scale_usd	scale_pre	排名时防止只由小基数增长驱动
post_trade_scale_usd	scale_post	经济规模
annual_growth_sd	growth_sd	年度增长波动
positive_growth_year_share	posyr_sh	post 期间正增长年份比例
rank_window_sensitivity	rank_sens	替代窗口下 rank 变化
source_bloc_sensitivity	bloc_sens	China-only 与 East-Asia source 定义差异
exposure_definition_sensitivity	expo_sens	rate/exposure/tariffed share 差异

8. M4 production-side 标准化变量

目标文件：production_validation_long.parquet。采用长表保存来源、指标与单位；禁止把不同概念合并相加。

8.1 通用 long schema

Canonical name	Stata name	定义
country_iso3	country_iso3	host/country
analysis_sector	ana_sector	共同 sector
year	year	年份
source_name	source	icio/amne/bea_mne/bea_fdi/mexico_fdi
metric_name	metric	指标名
value	value	原单位数值
unit	unit	current USD/current million USD/thousand persons/fraction
frequency	freq	annual/quarterly/H1
direction	direction	FDI inward/outward 或 affiliate trade direction
parent_country_iso3	parent_iso3	AMNE parent，适用时
direct_or_allocated	direct_alloc	direct/allocated/unavailable
suppressed_flag	suppressed	BEA
confidential_flag	confidentl	Mexico
aggregate_flag	is_agg	geography/sector aggregate
proxy_flag	is_proxy	coarse source-sector proxy
source_file	src_file	provenance

8.2 ICIO metrics

Metric	Stata metric label	Unit	Additivity/use
gross_output	gross_output	current million USD	production level
value_added	value_added	current million USD	production validation；允许 accounting negatives
exports	exports	current million USD	source-consistent export level
imports	imports	current million USD	source-consistent import level
intermediate_inputs	intermed_inputs	current million USD	IO intensity
taxes_less_subsidies	tax_less_sub	current million USD	accounting；允许负值
accounting_residual	acct_resid	current million USD	quality diagnostic

ICIO 只有 2017、2018、2022。禁止线性插值得到 2019 或 annual panel。

8.3 AMNE metrics

Metric	Unit	Direct/allocated	用途
affiliate_output	current million USD	direct bilateral	MP activity 主指标
affiliate_value_added	current million USD	allocated to parent	production validation
affiliate_exports	current million USD	allocated	affiliate trade
affiliate_imports	current million USD	allocated	affiliate input
foreign_affiliate_output_total	current million USD	host-sector total	foreign activity scale
parent_affiliate_output_share	fraction	derived direct output share	allocation provenance
affiliate_employment	unavailable	unavailable	PROHIBITED as zero/imputed

8.4 BEA MNE metrics

Metric	Unit	用途
employment	thousand employees	labor validation
sales	million current USD	affiliate activity
value_added	million current USD	production
capital_expenditures	million current USD	investment/activity
goods_exports_to_us	million current USD	affiliate -> U.S. direction
goods_imports_from_us	million current USD	U.S. -> affiliate direction

每个 metric 同时保留 suppression 和 small-value flag。aggregate geography 先排除。Transportation equipment 若作为 autos robustness，必须 proxy_flag=1。

8.5 BEA FDI metrics

Metric	Temporal type	Unit	规则
position	stock	million current USD	不跨年求和
flows	annual flow	million current USD	允许负 reversal
income	annual flow	million current USD	与 position/flow 分开

必须保留 direction。不能构造 total_fdi = position + flows + income。

8.6 Mexico FDI metrics

Metric	Stata name	规则
fdi_quarterly_usd	fdi_q_usd	quarterly flow，负值保留
fdi_annual_usd	fdi_a_usd	仅四季度完整且均非 confidential 时求和
fdi_h1_usd	fdi_h1_usd	2024 与历史 H1-to-H1 比较
fdi_complete_year_flag	fdi_yr_ok	2006-2023 full-year；2024 detail false
fdi_confidential_quarter_count	fdi_conf_n	confidential quarter count
fdi_disclosed_composition_share	fdi_discsh	只描述 disclosed cells，不能解释为完整 national share

Mexico national total 与 detailed composition 必须存为不同 source/metric，不能用 disclosed detail 替代全国 total。

9. M4-M5 production changes 和覆盖变量

对每个 source-metric 在其实际可用年份内构造变化，不能强迫所有来源使用同一连续窗口。

Canonical name	Stata name	公式/规则
value_level	value	原始 level
value_lag	value_lag	同 source-country-sector-metric 上一期
value_change	d_value	value-value_lag
value_dh_growth	gdh_value	非负 level 适用；signed FDI 不使用 DH level formula
value_log_growth	gln_value	两期均正时
signed_flow_change	d_flow	signed FDI flow 的 level change
post_pre_mean_change	d_postpre	在 source 可覆盖窗口内年度平均差
coverage_year_count	cover_yn	有效年份数
missing_value_count	miss_n	missing 数
suppressed_value_count	supp_n	suppressed 数
confidential_value_count	conf_n	confidential 数

M5 coverage flags:

Canonical name	Stata name
has_trade	has_trade
has_icio	has_icio
has_amne	has_amne
has_bea_mne	has_beamne
has_bea_fdi	has_beafdi
has_mexico_fdi	has_mexfdi
has_any_production	has_prod
has_direct_activity_measure	has_direct
has_only_financial_evidence	fin_only
source_count_available	source_n

has_only_financial_evidence=1 的 cell 不能标为 production confirmed。

10. M6 connector ranking

10.1 组件

推荐先把 raw components 转为同一比较组内的 percentile rank [0,1]，减少单位和极端值影响，同时永久保留 raw variables。

Component	Candidate raw inputs	经济含义
score_exposure	exposure overlap、sector tariff exposure	是否集中在受冲击产品
score_us_gain	connector-US level gain、DH growth	对美出口扩张
score_source_gain	China/East-Asia input gain、DH growth	source-to-connector link
score_replacement	China decline + connector gain、replacement ratio	U.S. import replacement
score_comovement	both/triple product share、weighted correlation	两条贸易腿同步性
score_scale	pre/post trade scale	经济规模
score_stability	positive-year share、negative volatility	时间稳定性
score_production	direct activity measures	生产侧验证

10.2 至少两种排名方案

方案 A，推荐主结果：透明 equal-weight percentile score。

```
rank_score_equal =
mean(score_exposure,
    score_us_gain,
    score_source_gain,
    score_replacement,
    score_comovement,
    score_scale,
    score_stability)
```

优点：透明、可复核、对单位不敏感。缺点：默认各维度同等重要。

方案 B，推荐稳健性：rerouting-core score，减少规模权重。

```
rank_score_rerouting =
0.20 exposure
+ 0.20 us_gain
+ 0.20 source_gain
+ 0.20 comovement
+ 0.10 replacement
+ 0.05 scale
+ 0.05 stability
```

优点：更贴近两段式 rerouting 机制。缺点：权重含研究判断，必须报告替代权重。

production evidence 不建议直接混入主 trade ranking；更清楚的做法是先得到 trade ranking，再给出 production_validation_grade。如果合成总分，必须作为第三套扩展结果。

10.3 Ranking outputs

Canonical name	Stata name	状态
rank_score_equal	score_eq	planned M6
rank_position_equal	rank_eq	planned M6
rank_score_rerouting	score_rr	planned M6
rank_position_rerouting	rank_rr	planned M6
rank_top3_equal	top3_eq	planned M6
rank_top3_rerouting	top3_rr	planned M6
rank_stability_across_specs	rank_stab	planned M6
rank_component_min	score_min	planned robustness
ranking_spec	rank_spec	planned M6

所有 rank 必须在包含 core connectors 和 comparators 的完整共同样本上计算，不能先删掉低排名观察。

11. M6 rerouting facts 和 production validation

11.1 Rerouting classification

Canonical name	Stata name	定义
china_us_decline_flag	cnus_dropf	China-US post < pre，且 level decline 达阈值
connector_us_gain_flag	conn_gainf	connector-US post > pre
source_input_gain_flag	src_gainf	China/East Asia -> connector post > pre
rerouting_two_leg_flag	rr_2leg	connector gain + source input gain
rerouting_three_condition_flag	rr_3cond	China decline + connector gain + source input gain
high_exposure_rerouting_flag	rr_hiexp	triple condition + high exposure
rerouting_evidence_grade	rr_grade	none/weak/moderate/strong，规则写入 config

Evidence grade 必须由预先固定的规则生成，不能根据结果手工改标签。

11.2 Production validation

Canonical name	Stata name	定义
output_growth_positive	out_pos	ICIO/direct output change >0
value_added_growth_positive	va_pos	VA change >0
affiliate_activity_growth_positive	aff_pos	AMNE/BEA activity >0
employment_growth_positive	emp_pos	available employment >0
capex_growth_positive	capex_pos	BEA capex >0
fdi_flow_increase	fdi_pos	signed flow change >0
direct_activity_signal_count	direct_n	output/VA/affiliate/employment/capex positive count
financial_signal_count	fin_n	FDI flow/position/income signals
production_confirmed_flag	prod_conf	至少一个 direct activity signal，不能只靠 FDI
financial_support_only_flag	fin_supp	只有 financial evidence
production_validation_grade	prod_grade	none/financial-only/partial/multiple-source

推荐 grade：

- none：无生产或金融支持；
- financial_only：只有 FDI；

- partial_direct：至少一个 direct activity metric 同向；
- multiple_direct：至少两个不同 source 或不同 direct activity families 同向；
- conflicting：trade 与 production signals 方向冲突。

12. Structural moments 和 Scenario A/B/C

必须区分“观测 moment”和“model shock/parameter”。数据不会直接观测 hgamma。

12.1 Trade-cost candidates

Canonical name	Stata name	公式	状态
gross_tariff_factor_hat	htau_gross	(1+tariff_rate_post)/(1+tariff_rate_pre)	structural target only
exposure_factor_hat	htau_expo	(1+exposure_post)/(1+exposure_pre)	robustness candidate
tariff_rate_change_pp	d_trf_pp	100*(rate_post-rate_pre)	descriptive moment
tariff_exposure_change_pp	d_exp_pp	100*(expo_post-expo_pre)	descriptive moment

最终 h_tau 选择必须与模型中 iceberg trade cost 或 tariff wedge 的定义一致，不能仅因为变量名相似就直接导入。

12.2 MP/production targets

Moment	观测来源	用途
foreign affiliate output share change	AMNE	incumbent MP-intensive adjustment
affiliate output growth	AMNE	Scenario B target
affiliate sales/VA/employment/capex growth	BEA MNE	direct activity validation
gross output/value added growth	ICIO	broader production response
FDI position/flow change	BEA/Mexico	Scenario C financial upper-bound support

Candidate names：

- mp_affiliate_output_share_change -> d_affshr
- mp_affiliate_output_dh_growth -> gdh_affout
- bea_affiliate_activity_index -> bea_affidx
- production_activity_index -> prod_idx
- fdi_financial_index -> fdi_idx

这些是 moments，不是 hgamma 本身。模型校准后才生成：

- hgamma_scenario_b
- hgamma_scenario_c

并必须记录 calibration mapping、normalization、elasticity、target source 和 solver result。

12.3 Structural moment table

目标字段：

Field	含义
moment_id	稳定唯一标识
scenario	A/B/C
country_iso3	country/connector
analysis_sector	sector
moment_name	名称
moment_value	数值
unit	单位

Field	含义
source_name	来源
sample_start, sample_end	样本期
formula_version	公式版本
direct_or_proxy	direct/proxy/allocated/financial
can_enter_model	是否允许进入模型
exclusion_reason	不允许时原因
validation_status	PASS/FAIL

13. Controls、fixed effects 和 robustness 变量

这些变量用于后续实证，不应与核心 outcome 混合。

Family	Variables
time	year、post2018、post2020、pre/shock/post indicators
connector FE	connector_iso3 categorical
sector FE	analysis_sector categorical
baseline scale	log pre trade level only when positive；pre level raw
baseline China dependence	sector China share 2017
baseline exposure	2017/2018 sector exposure
proxy/scope	TWN proxy、BEA autos proxy、medical unsupported
coverage	has source flags、missing/suppression/confidential counts
small base	pre_zero、small_base、log_defined
alternative windows	2014-2017 vs 2020-2023；exclude 2024；shock-only
source bloc	China-only vs extended East Asia
policy	rate vs exposure vs tariffed share

2024 可能同时受政策 review 和数据时间边界影响。所有包含 2024 的结果要有 exclude-2024 robustness。

14. 不允许的变量和解释

禁止项	原因
把 Census 15-origin 合计命名为 world imports	分母范围错误
把 Census quantity zero 当真实数量	source unavailable
把 suppression/confidential missing 填 0	改变官方含义
把 BEA position + flow + income 相加	stock/flow/概念不同
把 FDI increase 命名为 plant entry	financial evidence 不能证明实体进入
把 trade rerouting flag 命名为 production relocation	缺少 direct production evidence
把 allocated AMNE VA/exports/imports 称为直接 bilateral observations	来源口径错误
把 repeated China-US benchmark 跨 connector 求和	机械重复
把 rate/share/exposure 直接求和到 sector	应重算分子分母
给 many-to-many concordance 等权拆分且不披露	无经验依据且会伪精细
用 arbitrary epsilon 让 ratio/log 看似完整	隐藏 undefined denominator
将 Scenario C 称为 structurally identified entry margin	当前模型不是 endogenous fixed-cost entry model

15. Stata 文件、主键和验证合同

15.1 计划交付

DTA	Stata isid	用途
m2_product_treatment.dta	isid hs6 year	policy product panel
m2_connector_trade.dta	isid conn_iso3 hs6 year	product-level trade
m3_trade_sector_year.dta	isid conn_iso3 ana_sector year	sector-year trade
m3_trade_sector_window.dta	isid conn_iso3 ana_sector window_id	pre/shock/post
m4_production_long.dta	source-specific composite key	production evidence
m5_connector_evidence.dta	isid conn_iso3 ana_sector year	integrated interface
m6_connector_ranking.dta	isid conn_iso3 ana_sector rank_spec	ranking
m6_structural_moments.dta	isid moment_id scenario	model handoff

15.2 Storage types

- monetary levels：double；
- rates/shares/growth：double；
- year：int；
- month/quarter/flags：byte；
- counts：按范围使用 int 或 long；
- ISO3：str3；
- HS6：str6；
- sector/source/status：固定上限字符串，不使用易丢信息的过短类型。

15.3 必须运行的 Stata 检查

```
version 18
clear all
set more off

use "data/processed/stata/m3_trade_sector_year.dta", clear
describe
isid conn_iso3 ana_sector year
assert inrange(year, 2014, 2024)
assert cn_us_val >= 0 if !missing(cn_us_val)
assert cn_conn_val >= 0 if !missing(cn_conn_val)
assert ea_conn_val >= 0 if !missing(ea_conn_val)
assert conn_us_val >= 0 if !missing(conn_us_val)
assert inrange(sec_expo, 0, 1) if !missing(sec_expo)
assert qty_ok == 0 if !missing(qty_ok)

count
summarize cn_us_val cn_conn_val ea_conn_val conn_us_val sec_expo
display "DATA_HANDOFF_VALIDATION_SUCCESS"
```

最终 .do 要逐表检查 rows、keys、ranges、missing counts、totals 和 success marker，并把 log 写入项目 F 盘。

16. 每个变量完成后的登记模板

Canonical name:
Stata name:
Status:
Output file:
Grain and key:
Source fields:
Formula:
Window:
Unit:
Additivity:
Zero rule:
Missing rule:
Negative-value rule:
Suppression/confidentiality rule:
Denominator rule:
Small-base rule:
Winsorization/standardization:
Purpose:
Allowed interpretation:
Prohibited interpretation:
Validation checks:
Validation result:
Output hash:
Reviewer/date:

17. 当前施工顺序

1. M3.1 已构造 connector-sector-year levels、coverage 和 sector treatment；
2. M3.1 product-to-sector value reconciliation 与确定性检查已通过；
3. 下一步 M3.2 构造 window annual means；
4. M3.3 固定 zero/small-base/growth rules；
5. M3.4 构造 replacement、source input、exposure overlap 和 co-movement；
6. M3.5 保留所有 connectors/comparators；
7. M3.6 生成窗口、source bloc 和 exposure sensitivity；
8. M4 逐来源建立 production long；
9. M5 逐来源 left join 和 coverage；
10. M6 先 trade ranking，再 production grade，最后 structural moments；
11. M7 导出 Stata、运行 .do、冻结 release。

18. 版本记录

Version	Date	Change
v0.1.0	2026-07-27	建立 M2 已验证变量、M3-M6 planned variables、公式、用途、缺失规则、排名方案、structural moment 边界和 Stata handoff 合同。
v0.1.1	2026-07-27	与完整数据处理审计档案建立双向合同；规定 M3-M7 每步必须同步登记实际输出、validation、hash 和 Stata 接口。
v0.2.0	2026-07-28	完成 M3.1 变量合同：登记 1,001 行 connector-sector-year 输出、70 个 canonical/Stata aliases、实际公式、flow-specific coverage、17/11 个 treatment denominator 异常、non-additive flags、validation 和确定性输出 hash；M3.2-M3.6 保持 planned。

下一次强制更新：M3.2 完成时，补齐窗口年度均值、窗口覆盖、缺失规则和实际 output/validation/hash；同时更新审计档案并重建两份 PDF。